

Remote Sensing

Environmental Data-Science

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Courtesy: SUPARCO Islamabad

What is Remote Sensing?

Remote sensing involves using sensors to gather data about the Earth's surface or atmosphere from a **distance**, typically through satellites, aircraft, or drones.

Remote Sensing

Remote sensing is the process of collecting information about an object, area, or phenomenon without making physical contact with it.

It plays a crucial role in various fields such as environmental monitoring, agriculture, urban planning, and more.

How Sensors Works?

Sensors on these platforms capture electromagnetic radiation emitted or reflected by the Earth's surface.

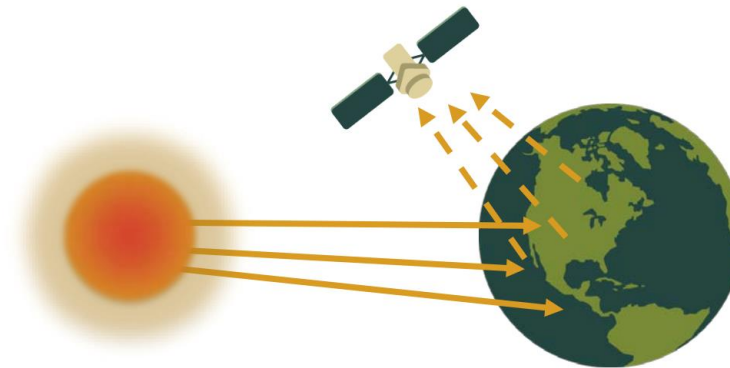
This data is then used to create images and maps.

Types of Remote Sensing

Passive Remote Sensing: Measures natural energy (sunlight) that is reflected or emitted by objects.

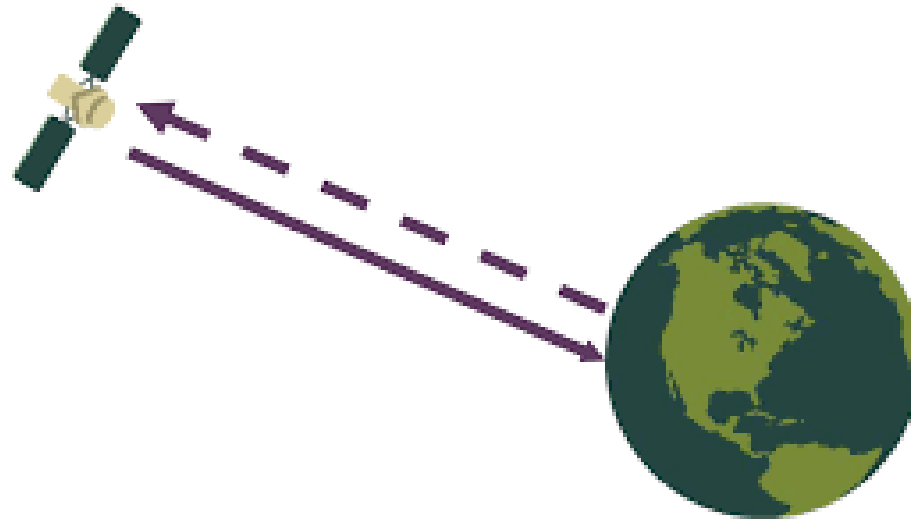
Examples include satellite imagery and photos.

Passive Sensors



Active Remote Sensing: It involves sending out energy (radar or laser) and measuring the reflected signal. Examples include radar and LiDAR.

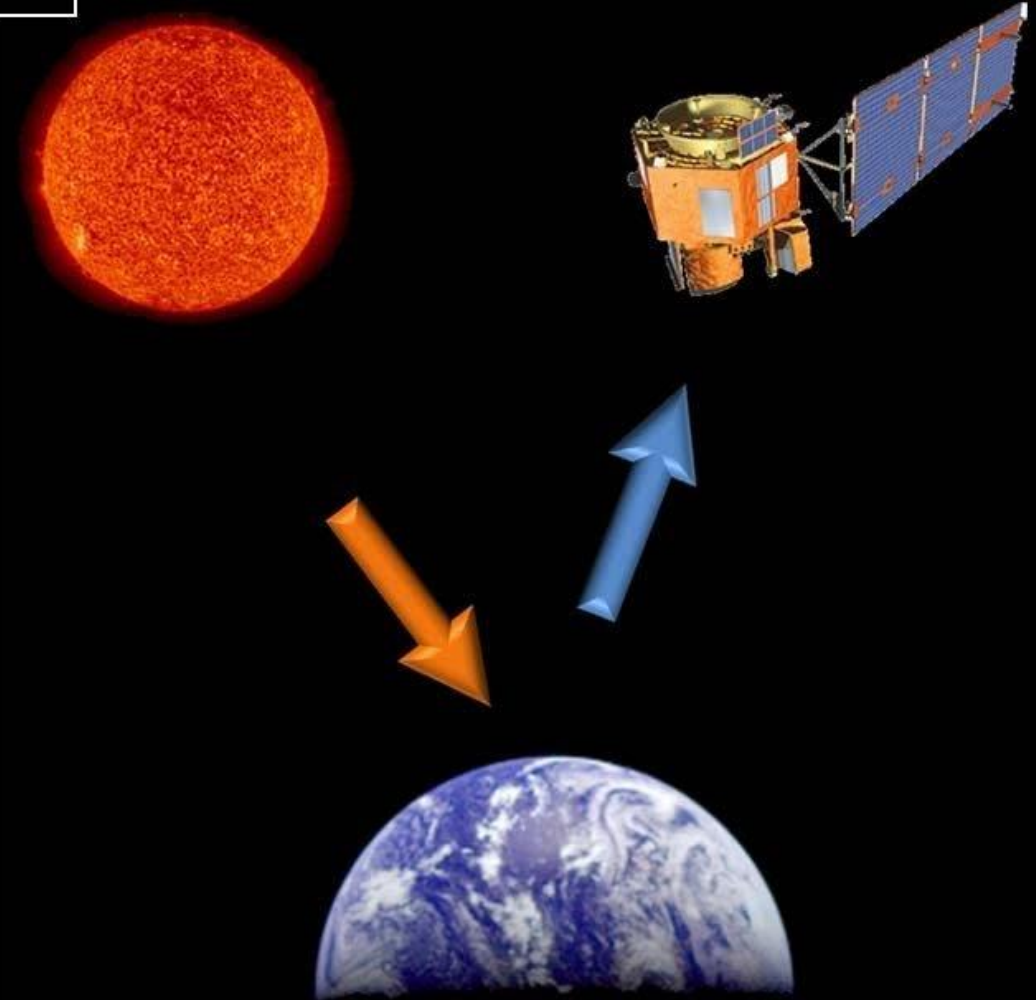
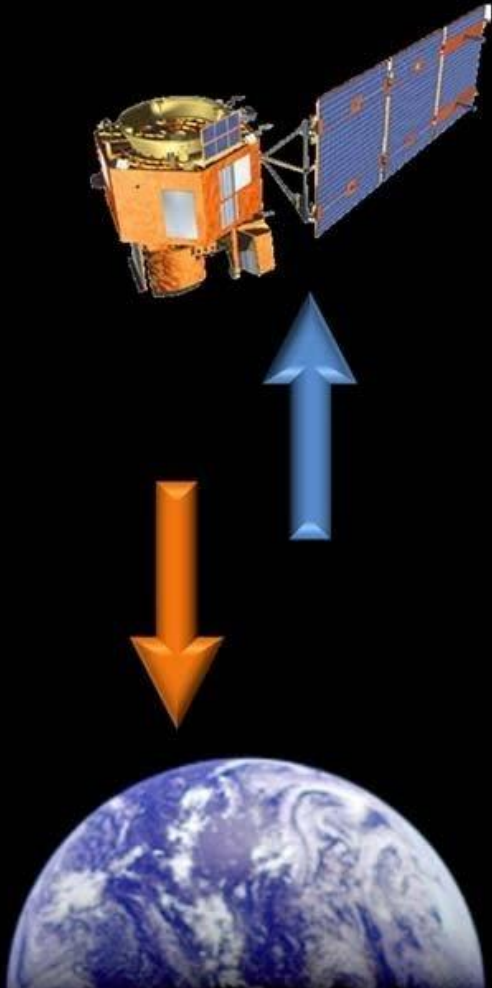
Active Sensors



ACTIVE REMOTE SENSING

VS

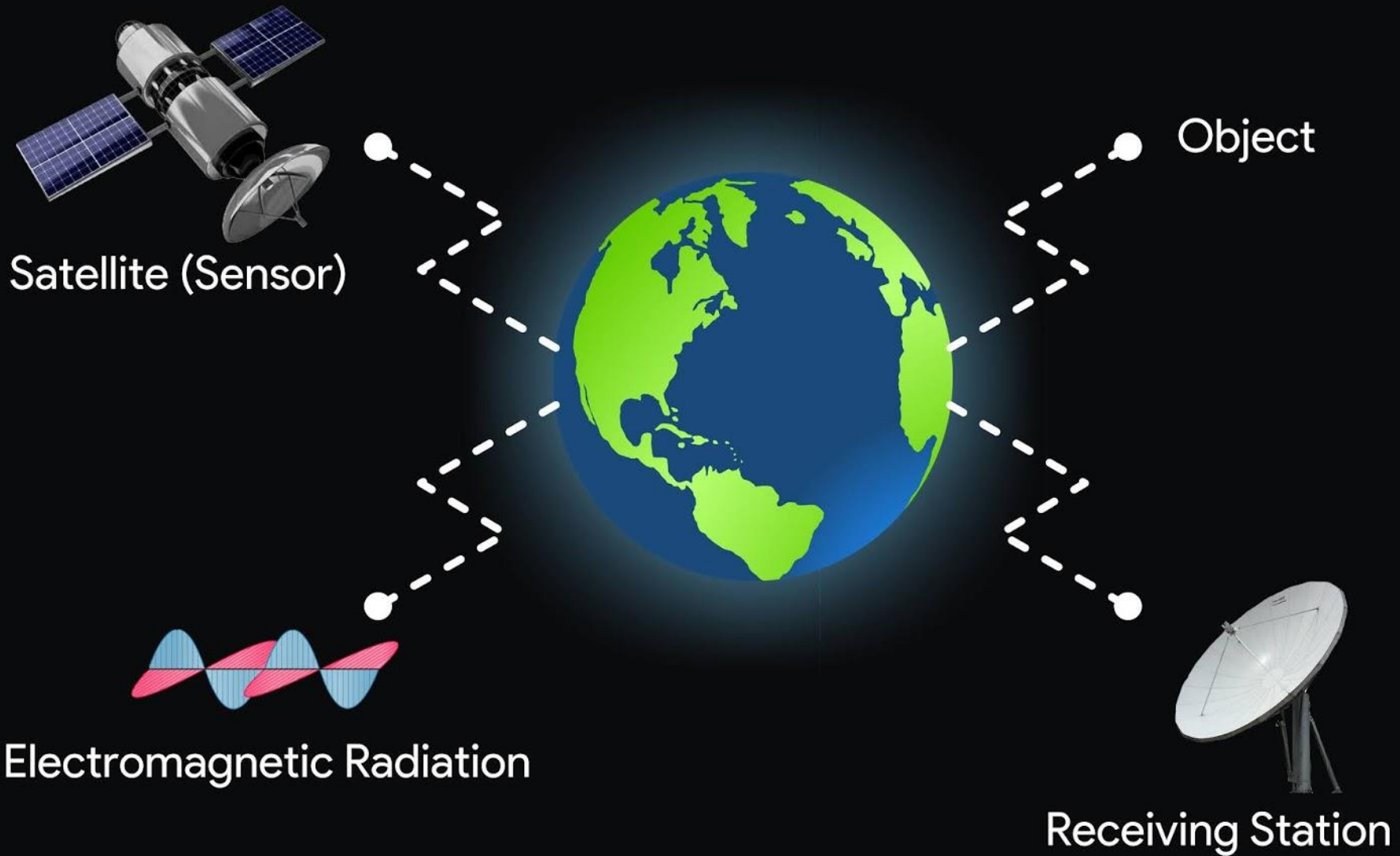
PASSIVE REMOTE SENSING



Remote Sensing Processes

1. Data acquisition (sensors capture data)
2. Data transmission (sending data to ground stations)
3. Data processing (converting data into usable images/maps)
4. Analysis (interpreting the data for insights).

Remote Sensing



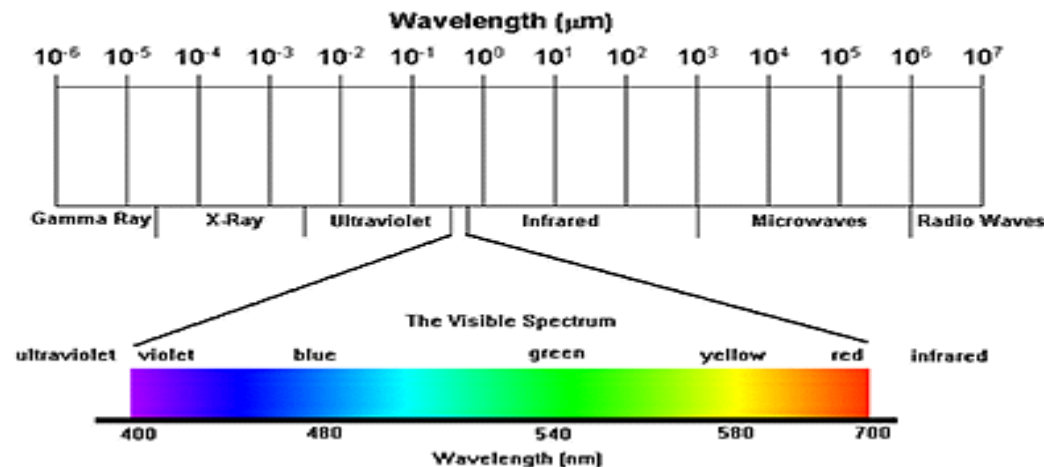
Remote Sensing Platforms

1. **Satellites:** Orbit Earth and capture data from space.
2. **Aircraft:** Flown over specific areas for more detailed data.
3. **Drones:** Small unmanned aircraft used for close-up observations.
4. **Ground-based systems:** Fixed sensors or cameras placed on the Earth's surface.

Electromagnetic Spectrum

Explanation of different parts of the spectrum used in remote sensing:

1. **Visible light**: Used for colour imagery.
2. **Infrared**: Detects heat and vegetation health.
3. **Microwave**: Penetrates clouds for all-weather observations.
3. **Radar and LiDAR**: Used for measuring elevation and creating 3D models.



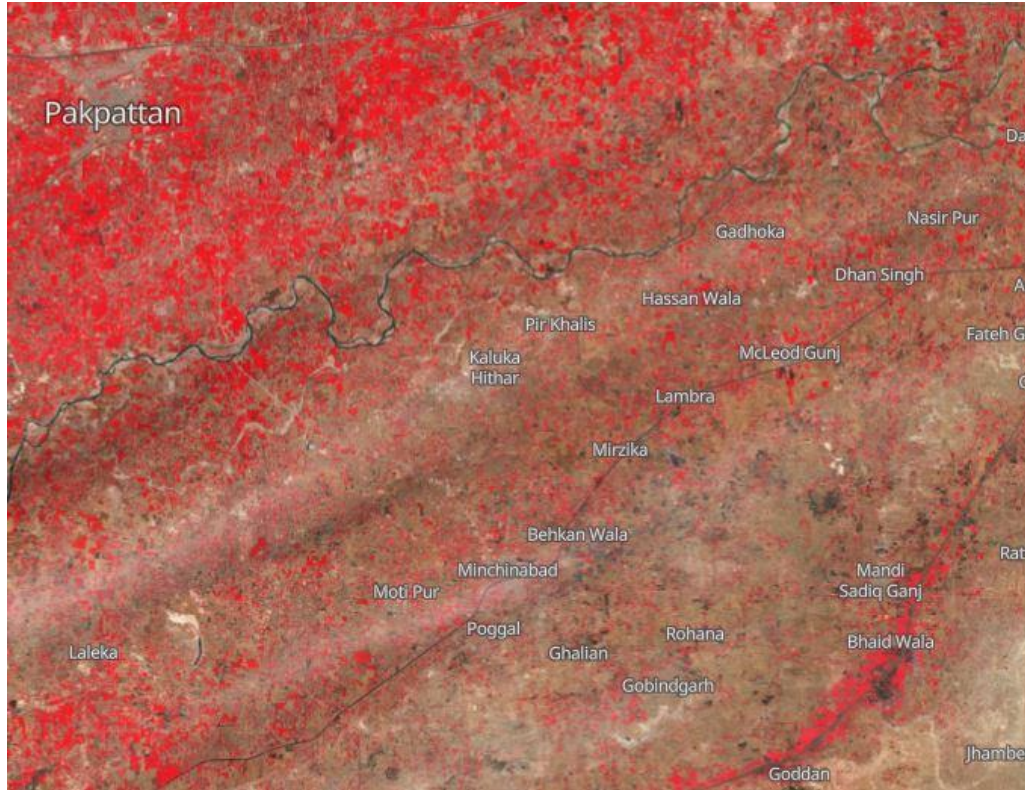
Application of Remote Sensing

1. **Agriculture**: Monitoring crop health, predicting yields, and optimizing irrigation.
2. **Environmental monitoring**: Tracking deforestation, analyzing pollution, and assessing ecosystem changes.
2. **Urban planning**: Mapping land use, managing infrastructure, and monitoring urban growth.
3. **Disaster management**: Assessing damage after events like floods, earthquakes, or wildfires and GLOF events and avalanches.
4. **Weather forecasting**: Gathering atmospheric data for accurate predictions.

Examples of Remote Sensing

Satellite Imagery for Flood Monitoring

25 April 2023



25 August 2023

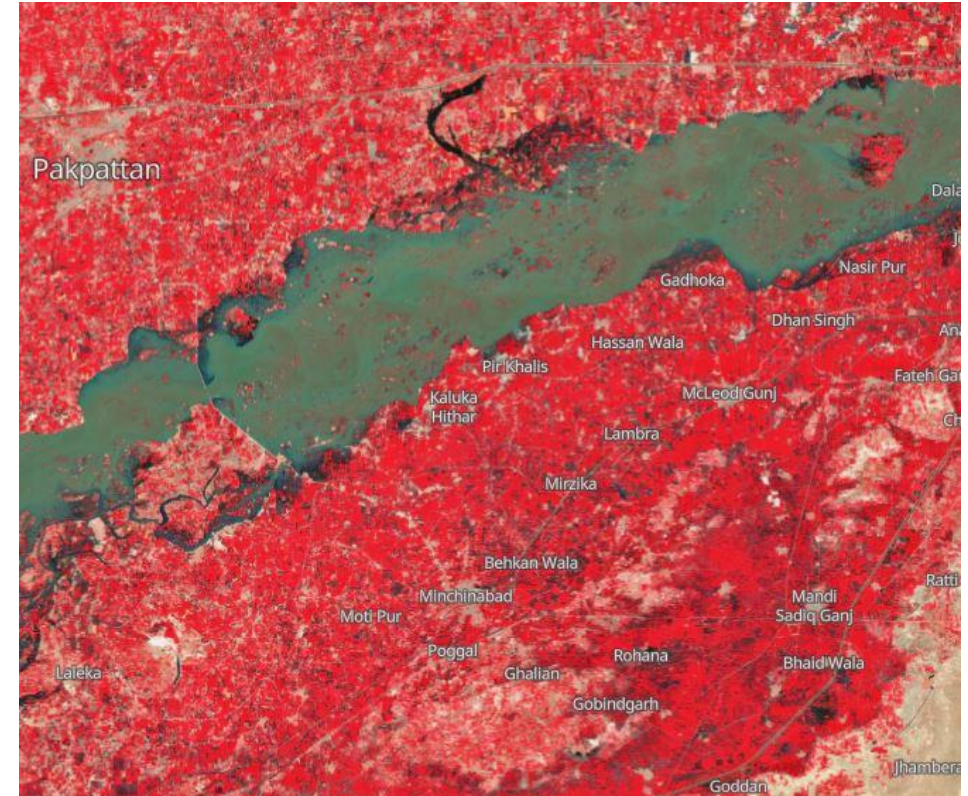
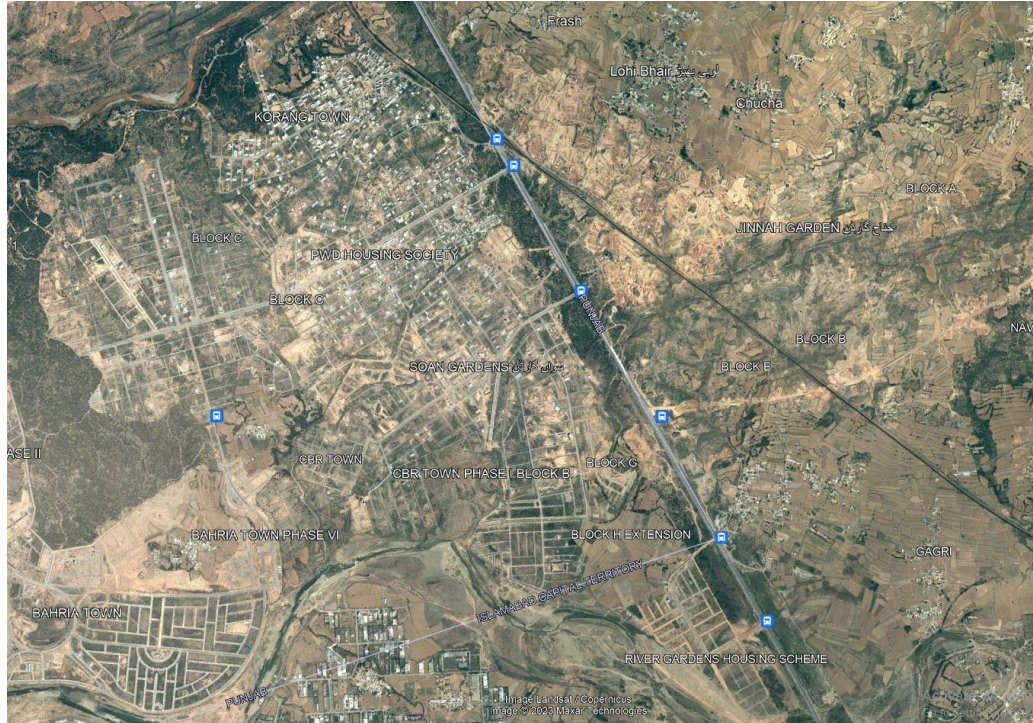


Image Source: Sentinel-2 Satellite

Satellite Imagery for Urban Sprawl

2002



2023

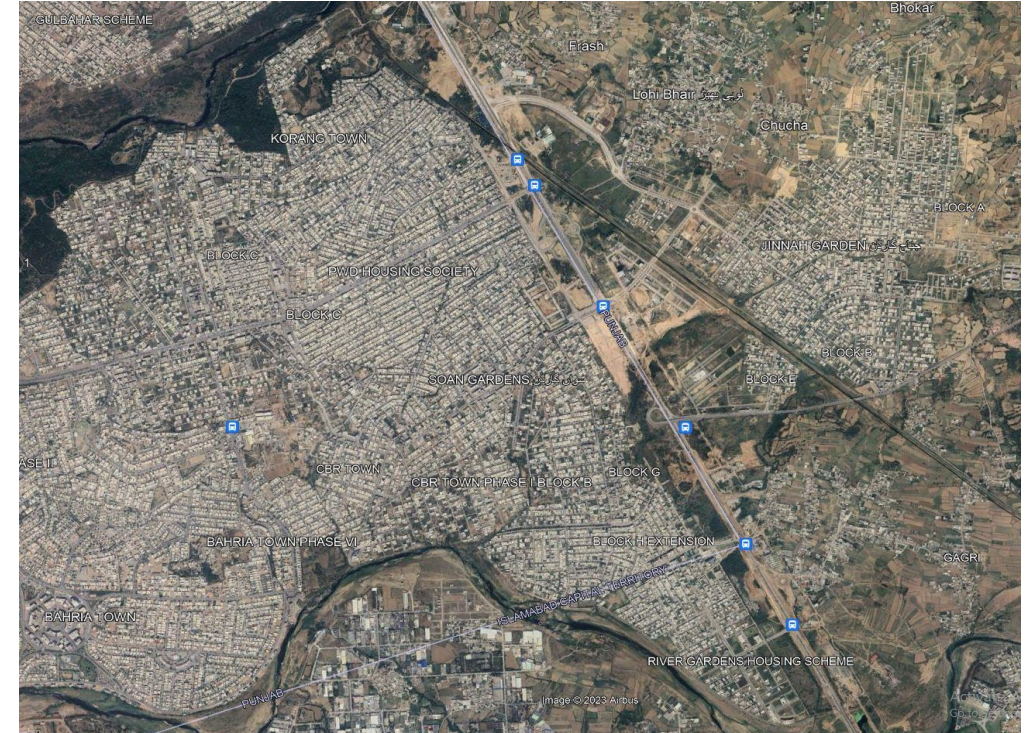


Image Source: Google Earth

Satellite Imagery for Forest Fire

Khyber Forest Fire 23 May 2022

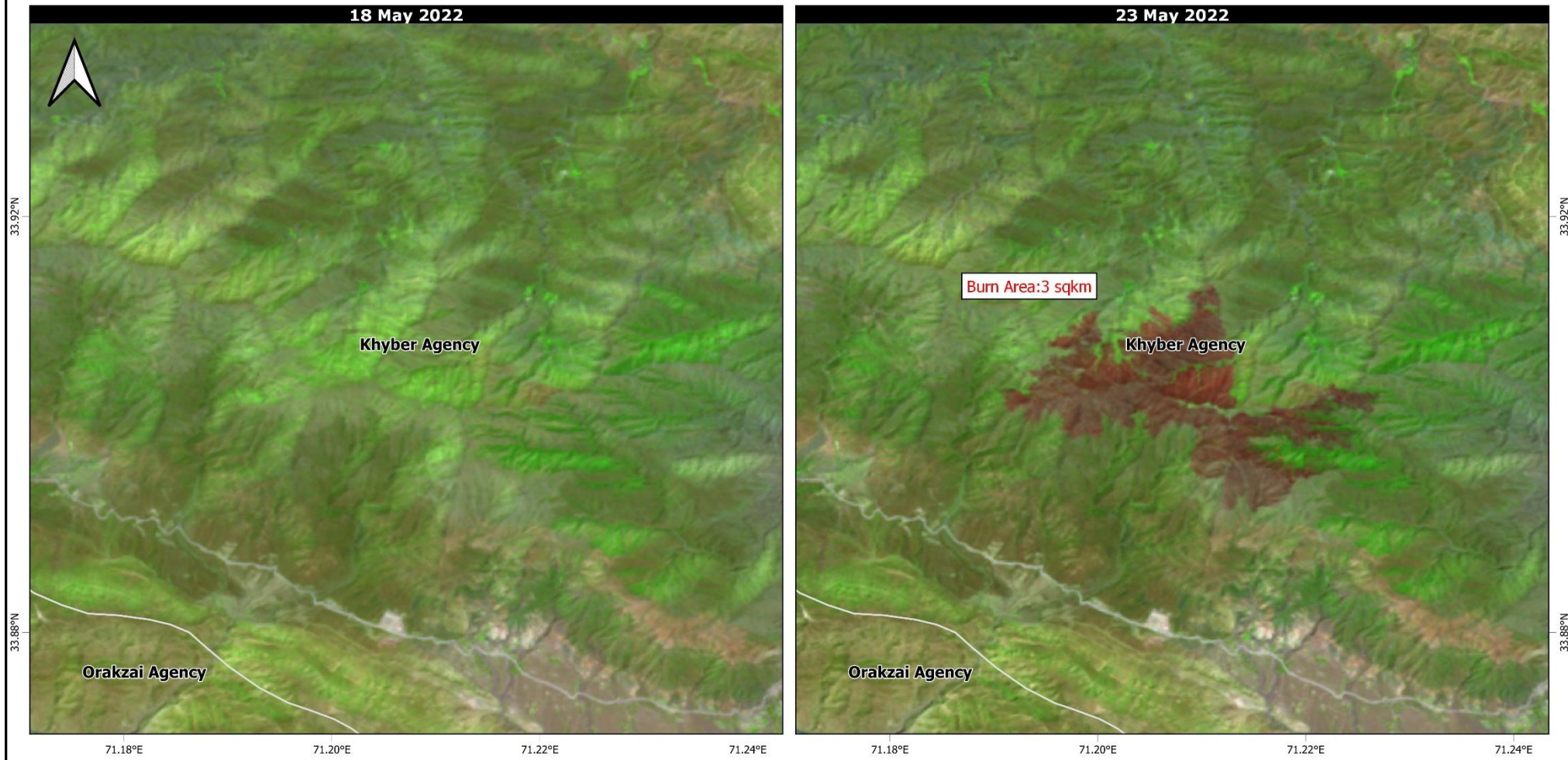


Image Source: Sentinel-2

There are different data analysis techniques using different software like ArcGIS, Qgis, etc.